**Dynamical Systems 1**
Course 01617
E25

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# Assignment 1

Note: Please include references when applying results from the textbook [Per00].

Problem 1:

Non-dimensionalization is a technique that can reduce the dimension of the space of parameters, thus making it easier to analyze a family of differential equations. In practice, one make a linear change of coordinates in each of the space variables, and in time, resulting in dimensionless new coordinates. If the change of variables are chosen well, the new system can often be made simpler than the original.

To be specific we shall consider the Duffing equation with damping and external harmonic driving according to

$$M\frac{d^2X}{dT^2} + \alpha\frac{dX}{dT} = kX - \kappa X^3 + A\sin(\Omega T)\ . \tag{1}$$

The Duffing equation governs the nonlinear oscillations of a mass $M$ attached to a spring exerting the force $f(X) = kX - \kappa X^3$ on the mass. The variable $X(T)$ denotes the position of the mass at time $T$. The term with the factor $\alpha$ denotes linear damping and the term $A\sin(\Omega T)$ is an oscillating harmonic force with amplitude $A$ and frequency $\Omega$. The unit of the position $X$ is meter and the unit of time $T$ is seconds. The Duffing equation also appears in many other physical contexts where nonlinearity plays a role.

The number of parameters in the Duffing equation can be reduced by introducing the linear scaling

$$T = k_t t \qquad \text{and} \qquad X(T) = k_x x(t)\ . \tag{2}$$

Here we shall choose the unit seconds on $k_t$ and the unit meter on $k_x$ leaving $t$ and $x(t)$ both non-dimensional. For numerical implementation of the Duffing equation this possesses the advantage, that we do not need to be concerned about units in the numerical code.

a) Show that by invoking the scaling in (2) into the Duffing equation (1) you can choose the scaling factors $k_t$ and $k_x$ such that the Duffing equation reduces to

$$\frac{d^2x}{dt^2} + \beta\frac{dx}{dt} = x - x^3 + a\sin(\omega t)\ . \tag{3}$$

In the process you shall determine $k_t$, $k_x$ and the new parameters $\beta$, $a$ and $\omega$.

b) Is it possible to choose $k_t$ and $k_x$ in a different way?

Problem 2:

We consider the first order differential equation

$$\frac{dx}{dt} = f(x) = (x+1)^2(1-x)\ , \tag{4}$$

where $x : \mathbb{R} \to \mathbb{R}$.

a) Draw a sketch of $f$ as function of $x$ and identify the stationary points for the differential equation (4).

b) Sketch the solution flow $\phi_t(x_0)$ for the equation (4) and determine the stability properties of the stationary points. Is it possible to determine the stability from $f'(x_0)$ for all stationary points $x_0$?

c) Sketch the solutions $\phi_t(x_0)$ as function of $t$.

d) Show that for $x \in ]-1;1[$ the differential equation (4) possesses the solution $x(t)$ given in the implicit form

$$\frac{1}{4}\ln\left(\frac{1+x}{1-x}\right) - \frac{1}{2}\frac{1}{x+1} = t - t_0 \ . \tag{5}$$

Here $t_0$ is an integration constant you can set to zero. Hint: Use separation of variables and decomposition of rational functions. Sketch the above solution $x$ as function of $t$.

e) Discuss the limits of $t$ as $x \to -1_+$ and as $x \to 1_-$ .

Problem 3:

A system of differential equations is given by

$$\frac{d\mathbf{x}}{dt} = \mathbf{B}\mathbf{x} \ , \quad \text{where} \quad \mathbf{B} = \begin{bmatrix} -1 & 2 \\ 0 & -1 \end{bmatrix} \quad \text{and} \quad \mathbf{x}(0) = \mathbf{x}_0 \ . \tag{6}$$

We have $\mathbf{x} : \mathbb{R} \to \mathbb{R}^2$ and $\mathbf{x}_0 = (x_{01}, x_{02})^T$.

a) Solve the problem in (6) by using exponential matrices.

b) Sketch the phase portrait of the system in (6) and classify the dynamics.

c) Are there any invariant sets? Are there any stable or unstable subspaces of the linear system? If yes, identify them.

# References

[Per00] Perko, L., Differential Equations and Dynamical Systems, Texts in Applied Mathematics 7, Third Edition, Springer-Verlag, New York, 2001.